Finite Deadline Third Derivative and Fourth-Order Remainder for a Quantum Capture Latch

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***Abstract*—A hash-bound finite-reference third-derivative readout and exact retained fourth-commutator identity lower the conditional capture-error ceiling from $123/2^{23}$ to $111/2^{23} < 2^{-16}$. The interval remains 16 ± 2^{-5} at $\beta = 5/16$; the next dyadic candidate fails the conservative remainder. Mixed bath terms are retained. A separate full-kernel Leibniz comparison identifies seven derivative-transfer slots, but their MQM continuum enclosures remain open. Numerical controls are same-agent evidence. Physical promotion is zero.**

***Index Terms*—quantum control, finite reservoirs, certified derivatives, open-system error accounting**

I. Fixed contract and checkpoint provenance

The parent point-curvature packet [1] establishes conditional cold curvature $< 2^{-14}$, finite thermal curvature $< 2^{-13}$, and error $< 123/2^{23}$ on 16 ± 2^{-5}. This addendum keeps $\Omega = 64$, $B = 16$, $\kappa = 8$, $\beta = 5/16$, $T = 16$, $h = g = 1$, $\lambda^2 = 4\sqrt{2}$, the source launch and retained error target 2^{-16}. Post-exposure engineering is disclosed; candidates 2^{-k}, $3 \le k \le 24$, and horizon $U = 129/8$ were frozen before the new readout. Widening was not an acceptance requirement.

The source checkpoint has 192 hot columns, each with $3 + 4M + \binom{M+1}{2} = 19299$ coordinates. It was propagated by 128 degree-48 Taylor steps and reused after verifying SHA-256[1]. These are mathematical modes and sector coordinates, not physical qubits. Reuse is not an independent propagation test.

II. Exact retained fourth identity

Write $H = H_B + K$, and introduce superoperators

$$A = \mathrm{ad}_K, \quad C = \mathrm{ad}_{[K,H_B]}, \quad D = \mathrm{ad}_{[[K,H_B],H_B]}, \tag{1}$$

with $E = \mathrm{ad}_{[[[K,H_B],H_B],H_B]}$. Bath-only outer commutators vanish under partial trace. Right multiplication by ad_{H_B} becomes the derivation $\delta R = [R, \mathrm{ad}_{H_B}]$, where $\delta A = C$, $\delta C = D$, $\delta D = E$. Starting from $R_1 = A$, the exact recurrence is $R_{n+1} = R_n A + \delta R_n$. Consequently,

$$\begin{aligned}\mathrm{Tr}_B\, \mathrm{ad}_H^4\, X = \mathrm{Tr}_B\, \big(&A^4 + 3CA^2 + 2ACA + A^2C \\ &+ 3DA + AD + 3C^2 + E\big)X.\end{aligned} \tag{2}$$

The terms $3C^2$, $3DA$, AD and E cannot be dropped.

[1]Checkpoint hash: `cb3dfc942103e5dba574179c14540d610 7b2c30c5c523a165a7ae87da1819f32`. The regenerable 55 MB checkpoint is excluded from the archive; parent replay code and the frozen readout are captured.

The source $Q \le 2$ bounds are $\|K\| < 14$, $\|[K, H_B]\| < 256/3$, and $\|[[K, H_B], H_B]\| < 4096/3$. Flat-band sixth-moment exactness at $M \ge 4$ gives

$$\|\delta^3 c\|^2 = \frac{128 B^6}{7\pi}. \tag{3}$$

The third mixed commutator has exchange norm at most $\sqrt{2}\|\delta^3 c\| = 65536/\sqrt{7\pi} < 16384$. For the finite response input $\|X\|_1 \le 2\nu$, (2) yields

$$\begin{aligned}\tfrac{1}{2}\|\Delta^{(4)}_{2,M}\|_1 &\le \nu\big(16 \cdot 14^4 + 48\tfrac{256}{3}14^2 \\ &\quad + 16\tfrac{4096}{3}14 + 12(\tfrac{256}{3})^2 + 32768\big) \\ &= 1843456\nu.\end{aligned} \tag{4}$$

The integer factorization is $1843456 = 2^8 \cdot 19 \cdot 379$, an arithmetic decomposition of this norm charge. It does not identify a geometric capacity law. The finite occupation trace ν is not a continuum photon-number bound.

III. Deadline third-derivative enclosure

For the evolved hot matrix W, let $A_j = H^j W$. The direct covariance-weighted readout is

$$i\,\mathrm{Tr}_B(A_3 N A_0^\dagger - 3A_2 N A_1^\dagger + 3A_1 N A_2^\dagger - A_0 N A_3^\dagger), \tag{5}$$

minus $\mathrm{Tr}\, N$ times the cold third derivative. The returned half trace norm is approximately 1.0870×10^{-8}, diagnostic only. Acceptance uses the exact rational half sum of absolute real and imaginary entries plus the error ledger.

Let $h = 64$, d_H be the inherited parameter mismatch, d_n the occupation ℓ_1 mismatch, and e the parent finite propagation allowance. The third-derivative correction is

$$\begin{aligned}J_{\mathrm{err}} = \nu_U\big[&8h^3(2e + e^2) \\ &+ 8d_H((h + d_H)^2 + h(h + d_H) + h^2)(1 + e)^2\big] \\ &+ 8h^3 d_n + 2^{-12}.\end{aligned} \tag{6}$$

The final arithmetic slot is verified against the parent 2^{28} operation budget with $\gamma = 2^{-25}/(1 - 2^{-25})$:

$$\begin{aligned}&32\gamma\nu_U 8(h + d_H)^3(1 + e)^2 \\ &+16\,2^{-43}\nu_U(h + d_H)^3(1 + e)^2 < 2^{-12}.\end{aligned} \tag{7}$$

This assumes the declared round-to-nearest model and source operation counts; independent machine-arithmetic review remains open. Combining the rational returned-entry charge and (6) gives a finite deadline ceiling approximately $0.009861917 < 2^{-6}$.

Using the inherited slope and curvature, the new thermal variation is

$$d_{2,\mathrm{new}}(w) = 2^{-20}w + 2^{-13}w^2/2 + 2^{-6}w^3/6 + 1843456\nu_U w^4/24. \quad (8)$$

All other cold, Gaussian and uniform continuum VALUE slots are re-evaluated exactly as in the parent on $U = 129/8$. Cross multiplication on 22 candidates selects $w = 2^{-5}$. The diagnostic sum is $1.312179523150286 \times 10^{-5}$, enclosed by

$$E(w) < 111/2^{23} < 2^{-16}. \quad (9)$$

The reporting numerator 111 was selected after evaluation; acceptance stayed fixed. The halfwidth did not improve. At 2^{-4} the majorant is about 2.7774×10^{-5}. Keeping every other slot unchanged would require a fourth-derivative bound below 0.098109, versus the present 19.78283. This is an engineering requirement, not an actual-error lower bound.

IV. Separate continuum derivative gate

No uniform real-time value bound is silently differentiated. Suppose the SAME finite and continuum retained response kernels satisfy

$$\Delta(t) = -\int_0^t ds \int_0^s du\, F(t,s,u), \quad (10)$$

with three continuous dominated derivatives. F must include the final propagator, noncommuting operators, covariance and vacuum expectation; scalar filter cancellation is insufficient. Leibniz yields

$$\begin{aligned}\Delta^{(3)}(t) = &-(3F_t + 2F_s + F_u)(t,t,t) \\ &- \int_0^t (3F_{tt} + 3F_{ts} + F_{ss})(t,t,u)\, du \\ &- \int_0^t ds \int_0^s du\, F_{ttt}(t,s,u). \end{aligned} \quad (11)$$

If ϵ_α uniformly bounds half the trace norm of each finite-minus-continuum derivative kernel, the transfer debit is at most

$$3\epsilon_t + 2\epsilon_s + \epsilon_u + T(3\epsilon_{tt} + 3\epsilon_{ts} + \epsilon_{ss}) + T^2\epsilon_{ttt}/2. \quad (12)$$

The common MQM kernel map, regularity and seven numerical slots are OPEN. The proposed next auxiliary budget 2^{-12} has units of model-time^{-3} and is distinct from the channel target. The current window still uses only uniform continuum values.

V. Controls, repair and scope

Six seeded fourth-commutator controls at $M = 4, 8$ agree within 5.483×10^{-12}; two third-derivative routes agree within 6.120×10^{-19}. Two inherited cold-curvature controls also pass. An exact word recurrence verifies (2); 64 exact monomial controls verify every boundary coefficient in (11). These do not certify an MQM continuum embedding or constitute external review.

An initial diagnostic reused the symbol for a mixed commutator as the eigensystem matrix, incorrectly reporting its norm as one. The fourth identity had already been checked before that overwrite; acceptance was unaffected. The receipt is preserved, the name repaired and all controls rerun. Fresh mixed norms are approximately 13975.134, below 16384.

Physical promotion remains zero. Permanent retention, bath replenishment, arbitrary-input diamond accuracy, native admission and hardware validation remain open. Next work should tighten cold third-derivative variation and response-local fourth remainders together, while independently closing the full-kernel derivative comparison.

References

[1] B. W. Mayes, "Deadline curvature enclosures and a wider quantum latch capture window," MQM point-curvature v0.1, repository artifact commit fd18ebf8c955351670eb26721dc8fe9895501aff, 2026. Directed parameter, launch and continuum dependencies are archived in that release.